

Title: Constructing a Pokémon World: Analyzing Language Across Types and Generations

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Goal Statement: We will analyze Pokémon descriptions to identify patterns in descriptive vocabulary across Pokémon types and generations to determine which types and generations cluster together and which words characterize these clusters; success will be contingent on the production of well-separated clusters (high silhouette scores) with semantically cohesive and interpretable word patterns (distinct relative word frequencies within versus outside each cluster).

Research Question: How do the descriptive characteristics of Pokémon differ across types and generations, and which types or generations exhibit similar descriptive language?

Narrative Paragraphs

Pokémon's 30-year-old universe, spanning over 1,028 species [1], feels like one cohesive world because of the consistent tone used in writing Pokémon species entries. We are analyzing the explicit descriptive vocabulary of Pokédex entries across all nine generations and 18 types to determine how developers used language to construct distinct fictional realities within and across games. This matters because that same cohesiveness of Pokémon's multi-generational franchise can help developers achieve something similar when building their own IPs. Furthermore, fans creating their own Pokémon species, either to share with the community or for their own fan games, could benefit from referencing this existing worldbuilding template.

Currently, Pokémon's mechanical data, like types, stats, and movesets, are well documented through APIs like PokeAPI, but the prose of Pokédex entries has not been studied to the same degree. Prior work [2] has used Pokédex text to predict a Pokémon's type with one-hot encoding, GloVe, and transformers, but that work treated the text only as input for a prediction task. Our project shifts the focus from prediction to understanding the text as a worldbuilding mechanism. By identifying patterns in vocabulary across Pokémon types and generations, we aim to show how language can be used to maintain story cohesion in a decade-spanning franchise.

Modeling Approach

We will use hierarchical clustering to identify patterns in the vocabulary of Pokémon species entries across different Pokémon types and generations. Specifically, we will examine relationships among Pokémon types and generations based on their descriptive vocabulary to identify shared linguistic characteristics. Hierarchical clustering is an “unsupervised machine learning” [3] approach that detects similar objects and clusters them based on their degree of dissimilarity. This approach is appropriate because it allows us to explore naturally occurring relationships in our data without assuming beforehand what groupings will emerge. Resulting dendrograms also allow examination of relationships at multiple levels, permitting flexibility in the number of clusters that emerge. For example, we may find that Fire and Electric types share similar descriptive patterns, while Ground and Rock types exhibit their own unique degrees of similarity. These clusters further enable us to directly compare types and generations by which specific linguistic dimensions contribute to their groupings.

References

- [1] "Pokémon (species)," *Bulbapedia, the community-driven Pokémon encyclopedia*. [Online]. Available: [https://bulbapedia.bulbagarden.net/wiki/Pok%C3%A9mon_\(species\)](https://bulbapedia.bulbagarden.net/wiki/Pok%C3%A9mon_(species)). [Accessed: Sep. 9, 2026].
- [2] R. Belfer, "Predicting Pokémon Type Using Pokédex Entries," *Medium*, Jun. 11, 2021. [Online]. Available: <https://medium.com/analytics-vidhya/predicting-pok%C3%A9mon-type-with-the-pok%C3%A9dex-7038754dc422>. [Accessed: Sep. 9, 2026].
- [3] J. Noble, "What is hierarchical clustering?," *IBM*, [Online]. Available: <https://www.ibm.com/think/topics/hierarchical-clustering>. [Accessed: Sep. 9, 2026].